# Experimental units and independence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

## Name the entity. Count repeated records. Preserve the group boundary.

Distinguish assignment and measurement units and audit a new-plant split without assuming independence from IDs.

**Keep one habit:** Unit contract → query/run ledger → native whole-plant picture → split/dependence probes → metadata audit → independent classroom assignment.

MAIN LESSON · SECTION 1

## Name the unit before counting the measurements

> Repeated views or runs do not become new query contexts or plant objects.

### Learning goals

I can explain the **mechanism** and terminology of **experimental unit**s and independence.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Retrieve repeated-case accounting and fair comparison design

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

## Prerequisite and starting response

- Recall Baselines and fair comparisons. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

**My starting explanation**

## End goal

Explain **experimental unit**s and independence using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

**Your learning target**

**Remember before reading**

This week you will practise:

Explain the **mechanism** and terminology of **experimental unit**s and independence

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

**Retrieve what you already know**

**This week’s end goal**

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MAIN LESSON · SECTION 2

# Distinguish assignment entities, sampled objects and observations

> The study question determines which grouping matters for the claim.

## Begin with three photos of one plant

Imagine photographing the same plant three times. You now have three images, but only one plant object. Name what receives a treatment in an experiment, what entity is sampled and what each measurement records; these need not be the same. In a paired query comparison, group repeated runs by query context.

Distinct IDs and disjoint train/evaluation groups are useful bookkeeping, but statistical independence still needs a justified design.

## Correctly define experimental and measurement units

An **experimental unit** is the entity assigned a treatment in the study design. Name sampling units and measurement units separately; repeated measurements from one unit may be related.

### Complete unit/count/split contract

| Unit/design item | Complete fictional lesson rule |
|---|---|
| Experimental unit | Entity assigned a treatment in the design; do not automatically equate it with each measurement |
| Sampling/measurement unit | Name separately according to how entities enter the study and observations are collected |
| Main repeated-query case | Q1, Q2, Q3 each have four run records:12 measurements across 3 query contexts |
| Paired A/B comparison | Match each query across methods; actual assignment/randomization design must still be stated |

| Unit/design item | Complete fictional lesson rule |
|---|---|
| Picture plant dataset | P1–P4 each generate three view records:12 photos, 4 plant IDs |
| Intended picture use | Assess new-plant behavior, so a plant ID cannot occur in both training and evaluation |
| Declared split | Train all P1/P2 views; evaluate all P3/P4 views:6 photos in each |
| Independence limit | Neither distinct IDs nor this group-disjoint split proves statistically independent sampling |
| Scope | Synthetic metadata only: no actual images, model training, classroom assignment or student response collection |

### Name supplied and missing evidence

> **List the given facts and one fact that is not supplied.**

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

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MAIN LESSON · SECTION 3

## Trace twelve run records back to three query contexts

> A repeated measurement count cannot substitute for a new-context sample count.

## Original within-query pairing and run-variation statement

If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.

### Complete context-by-run ledger

| Main query context | Recorded runs | Measurements added | Distinct context count |
|---|---|---|---|
| Q1 | Runs 1–4 | 4 | 1 |
| Q2 | Runs 1–4 | 4 | 1 |
| Q3 | Runs 1–4 | 4 | 1 |
| Total | Three groups of four | 12 | 3 |
| Claim of 12 independent queries | Confuses repeated observations with contexts | Unsupported | Dependence/design assumptions still need stating |
| Paired-method extension | Match A/B on each query | Do not invent how many records per method were supplied | Query context remains grouping for that comparison |

## State the assignment/sampling design still missing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

**Experimental units and independence — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Three queries each run four times produce 12 outputs. | If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts. | There are three query units and 12 run measurements. Do not analyse them as 12 independent queries. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

**Connect the picture and the explanation**

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MAIN LESSON · SECTION 4

# Predict a plant view crossing the split boundary

> A unique image ID does not make its plant unseen.

## Current task and response

- Three queries each run four times produce 12 **output**s. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Predict moving one P1 view into evaluation

- Change: Put one P1 view into evaluation while other P1 views train.

▼ Check after predicting

**Prediction**: Flag object-level leakage for this new-plant target.

Reason: Evaluation now includes a plant already represented during **training**.

▼ Earlier wording, alternatives and picture transcripts

**Compare a familiar case with a changed case**

**Change one thing in the pictured case**

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MAIN LESSON · SECTION 6

## Read plant identity, view count and split membership together

> Do not call an already-seen plant unseen because one photo has a new filename.

### Main three-query case and checked counting

Three queries each run four times produce 12 **output**s.

There are three query units and 12 run measurements. Do not analyse them as 12 independent queries.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Separate four-plant inputs and design assumptions

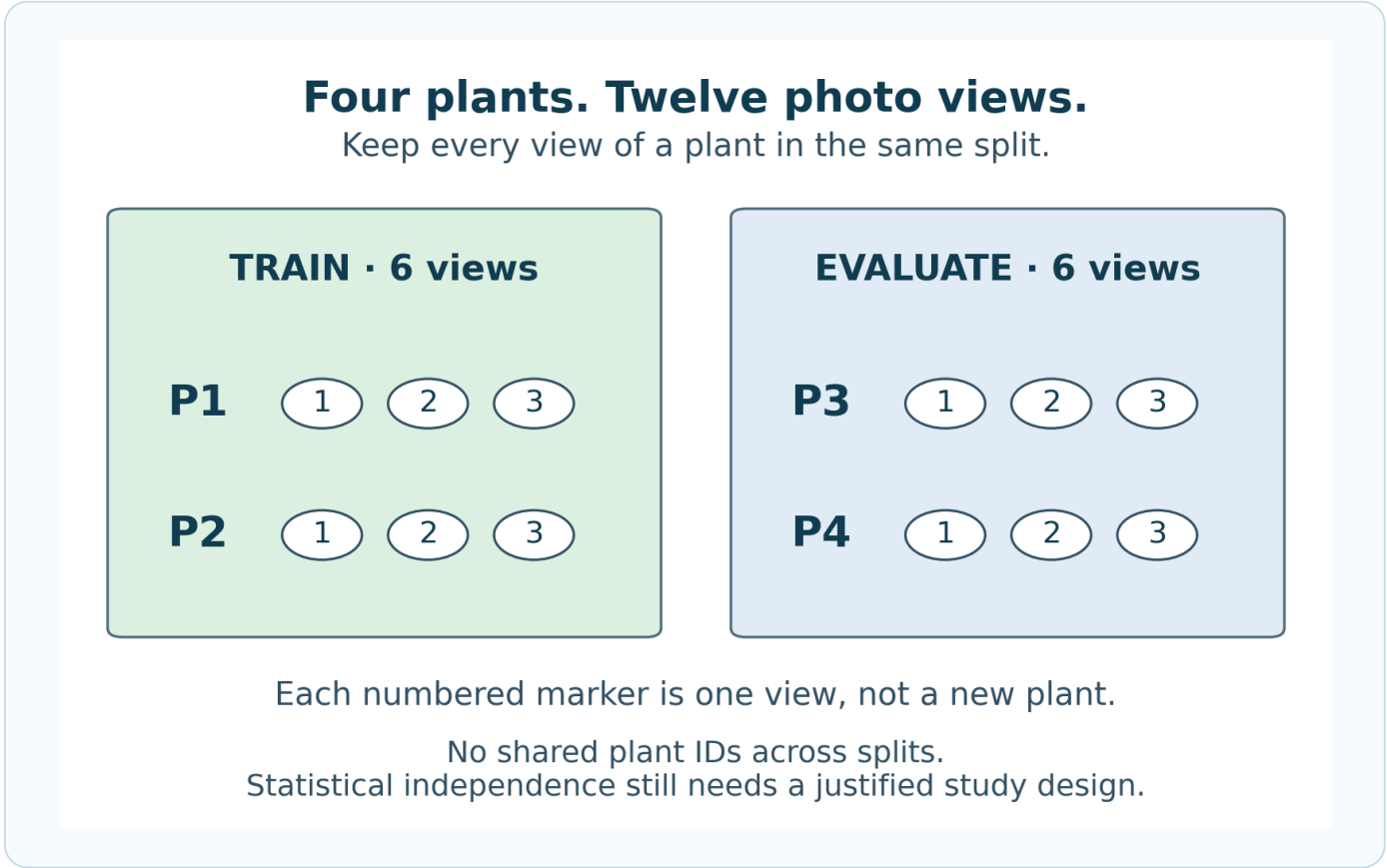
Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

Plant IDs P1–P4 each have three views. All views of a plant must stay in one split.

The picture is a separate plant-ID dataset. Four plant objects P1–P4 each have three views. For a new-plant goal, put all P1/P2 views in training and P3/P4 in evaluation:6 photos/2 plants in each, no shared plant IDs. Move P1-v1 to evaluation while P1-v2/v3 remain in training, and P1 now spans both sets.

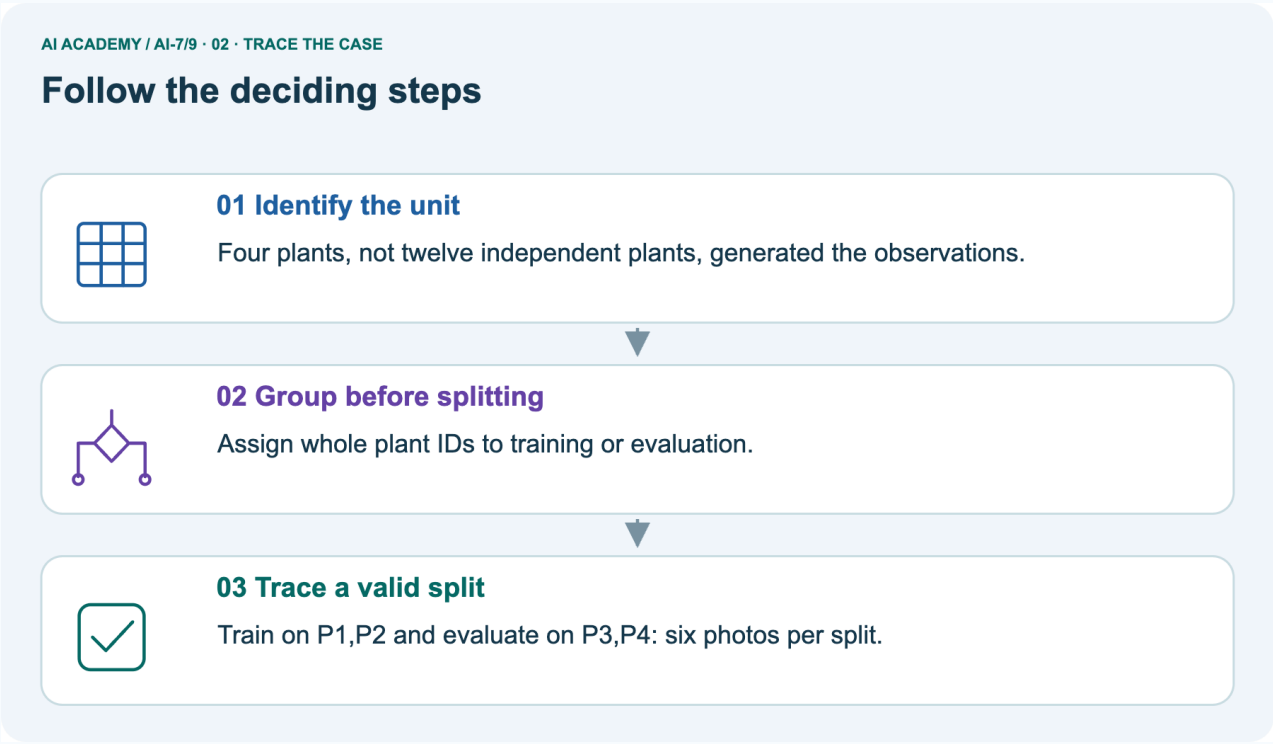
Counts become 5 training views and 7 evaluation views, but the main problem is shared object identity, not unequal split sizes. This is leakage for the declared new-plant goal. Four distinct plants may still share environmental/sampling dependencies; the metadata split alone cannot establish independence or broad coverage.

## Trace whole-plant groups with the native diagram



AI ACADEMY / AI-7/9 · 02 · TRACE THE CASE

## Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Identify the unit | Four plants, not twelve independent plants, generated the observations. |
| 2. Group before splitting | Assign whole plant IDs to training or evaluation. |
| 3. Trace a valid split | Train on P1,P2 and evaluate on P3,P4: six photos per split. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Repeated views from one object share information. Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

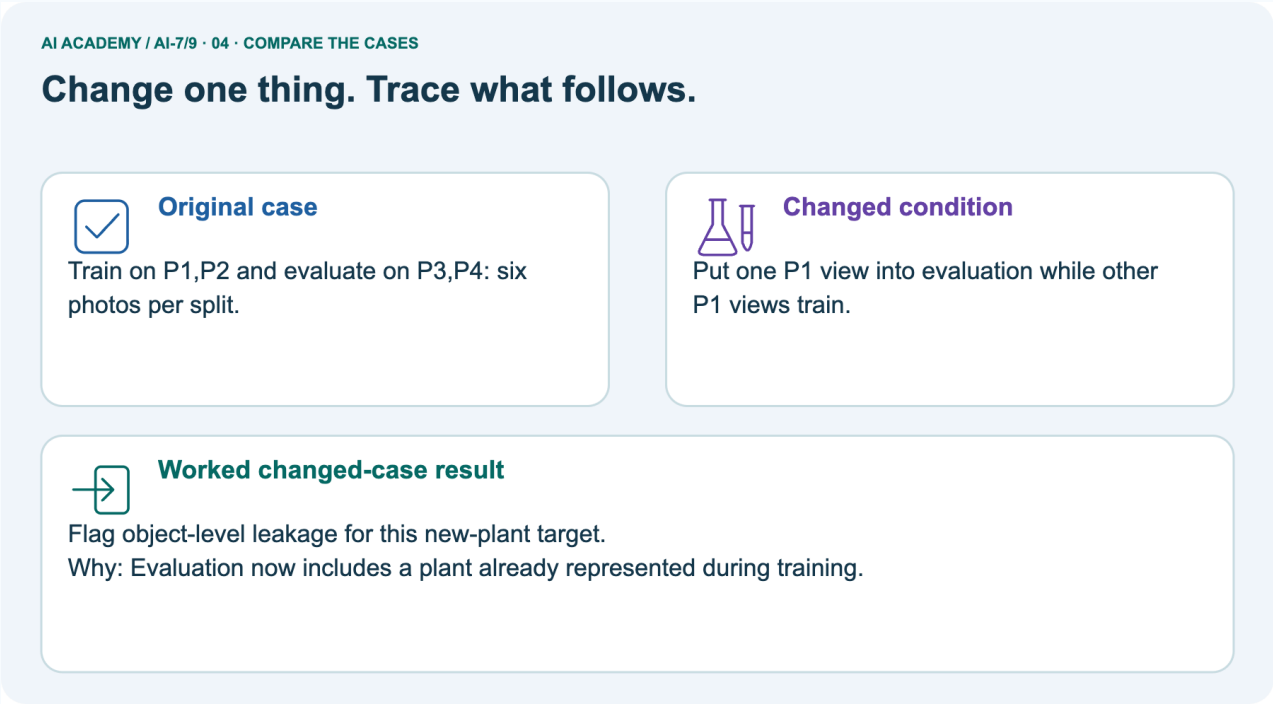
### Complete group-correct versus leaked-view counts

| Separate picture state | Training | Evaluation | Shared plant IDs |
|---|---|---|---|
| Group-correct split | P1/P2:6 views | P3/P4:6 views | None |
| Move P1-v1 to evaluation | P1/P2:5 views | P1/P3/P4:7 views | P1 |
| Across either complete dataset | 12 total view records | 4 distinct plant IDs | Counts alone do not establish independence |
| Interpretation | Whole-plant grouping protects this split boundary | Four plants offer limited population coverage | No classifier performance is measured |

## Predict P1 in both splits

- Change one thing: Put one P1 view into evaluation while other P1 views train. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting




Follow the labelled picture with the matching explanation. Open the vector diagram

## Explain small coverage and unverified independence

- Four plants offer little population coverage; even a group-correct split may be too small for broad conclusions.
- Read the labelled picture: Count distinct plant units and photo views, and state the independence assumption. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

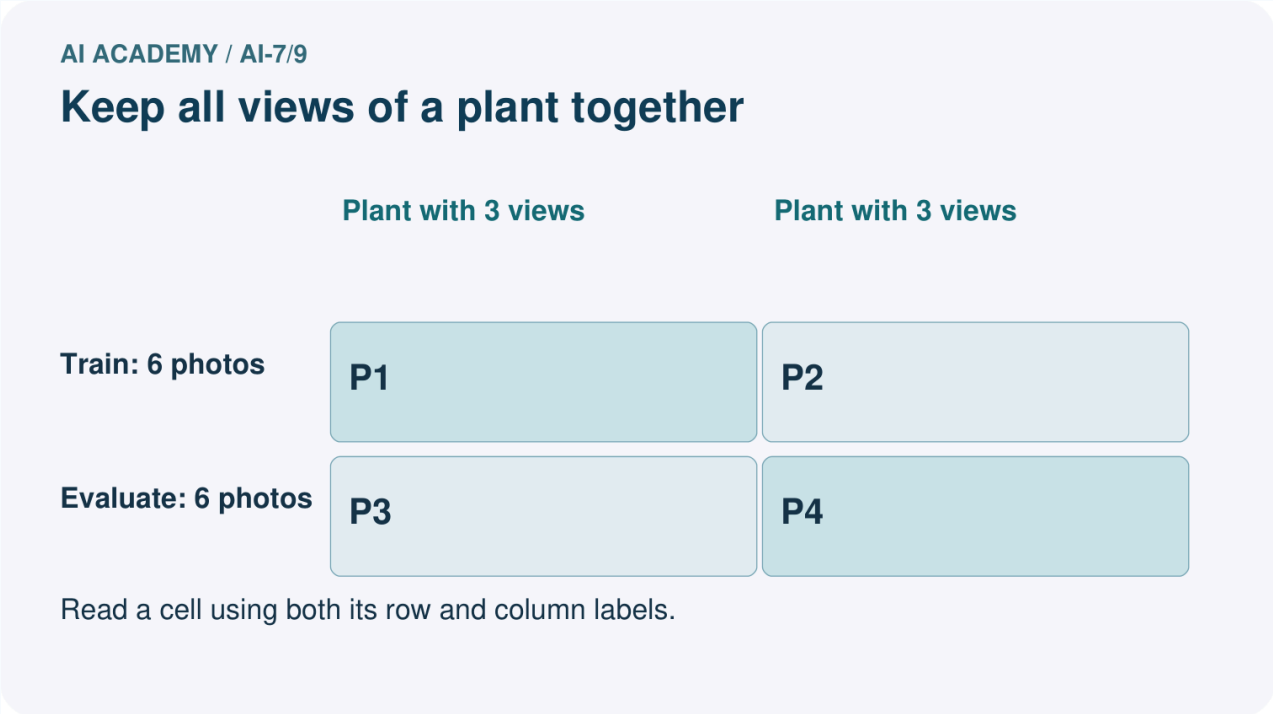
▼ Earlier wording, alternatives and picture transcripts

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

## A pictorial worked case: Experimental units and independence

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram




Follow the labelled picture with the matching explanation. Open the vector diagram

**Why this result follows**

**What this example does not establish**

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MAIN LESSON · SECTION 7

## Audit assignment, repeated measurement and leakage as different issues

> A sound count identifies the entity and the assumption, not just the number of rows.

## Core unit qualification and failure analysis

Independence follows design and relationships, not row count.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

### Complete grouping and dependence probes

| Unit/independence probe | Expected interpretation |
|---|---|
| Four runs for one query | Four measurements, one query context |
| Three views of one plant | Three photos, one plant identity |
| Distinct filenames for the same plant | Do not establish independent plant samples |
| No plant ID shared across splits | Passes that group-disjointness check only |
| Same plant spans train/evaluation | Leaks object identity for the declared new-plant goal |
| Independent treatment applied to whole groups | Assignment units are groups; measurements within them remain clustered |
| Query method comparison within each context | Preserve pairing; actual randomization/sampling still needs specification |
| A more balanced photo count | Cannot repair shared plant identity by itself |
| Group-correct tiny dataset | No broad new-plant reliability or population independence guarantee |

## Guided repair and explanation

- A report omits this qualification: Independence follows design and relationships, not row count. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

**Locate the earliest unsupported step**

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MAIN LESSON · SECTION 8

## Practise drawing the entity-to-measurement relationship

> Name what is repeated and which identity the split must preserve.

### Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

## Supported W2 trace

Use workbook W2 for the connected example "**Experimental unit**s and independence". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Individual scope check

- Use the worked case for **Experimental unit**s and independence. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

## Picture response

- Count distinct plant units and photo views, and state the independence assumption. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

**My explanation using a pictured step and the given facts**

| | |
|---|---|
| | |
| | |
| | |
| | |

### Reserve the selected classroom-level assignment case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

**Practise with less help: Trace the worked case**

**Check your reasoning**

**Explain the picture**

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MAIN LESSON · SECTION 9

## Run the complete unit-accounting and object-group split investigation

> Inspect both record identities and the stated inference boundary.

### Current repeated-query count investigation

Three queries each run four times produce 12 **output**s. Show the intermediate steps and give the conclusion with its scope.

### Runnable metadata counting and split audit

```
"""Count supplied query runs and plant groups; no experiment or model training."""
from collections import Counter
from copy import deepcopy
query_runs=[{'query_id':'Q'+str(q),'run':r} for q in range(1,4) for r in range(1,5)]
assert len(query_runs)==12 and len({row['query_id'] for row in query_runs})==3
views=[{'view_id':f'P{p}-v{v}','plant_id':f'P{p}'} for p in range(1,5) for v in range(1,4)]
before=deepcopy(views)
train_ids={'P1','P2'};eval_ids={'P3','P4'}
```

```
train=[row for row in views if row['plant_id'] in train_ids]
evaluation=[row for row in views if row['plant_id'] in eval_ids]

def audit_split(train,evaluation):
 for rows in (train,evaluation):
  if type(rows) is not list or not rows:raise ValueError('two nonempty view-record lists required')
  for row in rows:
   if type(row) is not dict or set(row)!={'view_id','plant_id'} or any(type(v) is not str or not v
for v in row.values()):
    raise ValueError('exact nonempty view/plant metadata required')
 all_rows=train+evaluation
 if not all_rows or len({r['view_id'] for r in all_rows})!=len(all_rows):
  raise ValueError('nonempty unique view identities across split required')
 train_plants={r['plant_id'] for r in train};eval_plants={r['plant_id'] for r in evaluation}
 return
{'train_views':len(train),'eval_views':len(evaluation),'distinct_plants':len(train_plants|eval_plants),
         'train_plants':sorted(train_plants),'eval_plants':sorted(eval_plants),
         'shared_plants':sorted(train_plants&eval_plants),
         'disjoint_for_new_plant_goal':not bool(train_plants&eval_plants),
         'independence':'not established from IDs alone'}
valid=audit_split(train,evaluation)
assert valid['train_views']==valid['eval_views']==6 and valid['shared_plants']==[]
# Move one actual view, rather than duplicate it: plant identity still crosses the boundary.
moved=next(r for r in train if r['view_id']=='P1-v1')
leaky_train=[r for r in train if r['view_id']!='P1-v1']
leaky_eval=evaluation+[moved]
leaky=audit_split(leaky_train,leaky_eval)
assert leaky['train_views']==5 and leaky['eval_views']==7 and leaky['shared_plants']==['P1']
assert not leaky['disjoint_for_new_plant_goal'] and views==before
print('Main query measurements:',len(query_runs),'distinct query contexts:',len({r['query_id'] for r
in query_runs}))
print('Picture four plant IDs, views per plant:',dict(Counter(r['plant_id'] for r in views)))
print('Group-correct split:',valid)
print('One P1 view moved into evaluation:',leaky)
print('No actual images, classifier runs, randomized assignments or human responses are used.
Disjoint IDs check leakage for this declared goal, not independence, representativeness or broad
new-plant reliability.')
```

## Expected counts and design/model limits

1. Predict 12 query-run records/3 contexts and 12 plant-view records/4 plant IDs. The local code builds metadata, not images. Its valid plant split has 6/6 views and no shared plant IDs. Moving one P1 view rather than duplicating it gives 5/7 views and shared P1, which fails disjointness for the new-plant goal. Original metadata remain unchanged.

   A disjointness flag does not test statistical independence, assignment randomization, sampling representativeness or model accuracy. No actual participant, classroom, plant collection or training run occurs; retain actual local output or mark paper-only trace not run.

## Evidence ledger and reflection

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

### ▼ Earlier wording, alternatives and picture transcripts

**Predict, investigate and preserve the evidence**

**Plan → try → check**

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MAIN LESSON · SECTION 14

# Try alone: group-level classroom assignment and individual responses

> Separate treatment assignment units from the measurements they generate.

### Complete selected independent task

Five classrooms each contribute 20 student responses under a classroom-level assignment. Identify assignment units and measurements.

## First attempt and support record

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

### Independent case: Apply a new case independently

### Attempt this without hints

Five classrooms each contribute 20 student responses under a classroom-level assignment. Identify assignment units and measurements.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

# Save assignment/sampling units, measurement IDs and split groups

> Keep design assumptions beside the record counts.

## Project task and artifact

Add a labelled **experimental unit**s and independence **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **experimental unit**s and independence.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

## Actual metadata run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

**Evidence for your project**

# Explain which entity was counted and what independence remains assumed

> A record is an observation, not automatically a new independent unit.

## Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **experimental unit**s and independence

**My deciding step, evidence and one remaining question**

## Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

## Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

### Vocabulary and your explanation

| Word | Meaning |
| --- | --- |
| Experimental unit | The entity independently sampled or assigned |
| Repeated measure | Another observation on the same unit |
| Clustering | Related measurements grouped within an entity |

**Use two terms to explain a step in this week's example.**

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

## Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

### Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.
- A report omits this qualification: Independence follows design and relationships, not row count. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.
- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

#### Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the research reasoning and its limits**

REFERENCE · SECTION 12

## Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

## Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

**My limited claim and a useful next check**

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

# Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

## Complete trace and evidence

### Complete evidence record

#### Experimental units and independence — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Three queries each run four times produce 12 outputs. | If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts. | There are three query units and 12 run measurements. Do not analyse them as 12 independent queries. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

#### My complete trace of the worked case

▼ **Reference · The earlier route checklist**

## Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

### Targets,route and first prediction

Start with this question: Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of experimental units and independence | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

## Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

### First supported question and optional hint

- Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

- Which labelled step matches this detail? Train on P1,P2 and evaluate on P3,P4: six photos per split.
- 1. Group before splitting
- 2. Identify the unit
- 3. Trace a valid split
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Trace a valid split Train on P1,P2 and evaluate on P3,P4: six photos per split. Repeated views from one object share information.

Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

## Changed-case question and optional hint

- Changed-case practice: Put one P1 view into evaluation while other P1 views train. Which conclusion is supported by the supplied facts?
- 1. Flag object-level leakage for this new-plant target.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Flag object-level leakage for this new-plant target. Evaluation now includes a plant already represented during **training**.

## Repair and retain the record

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

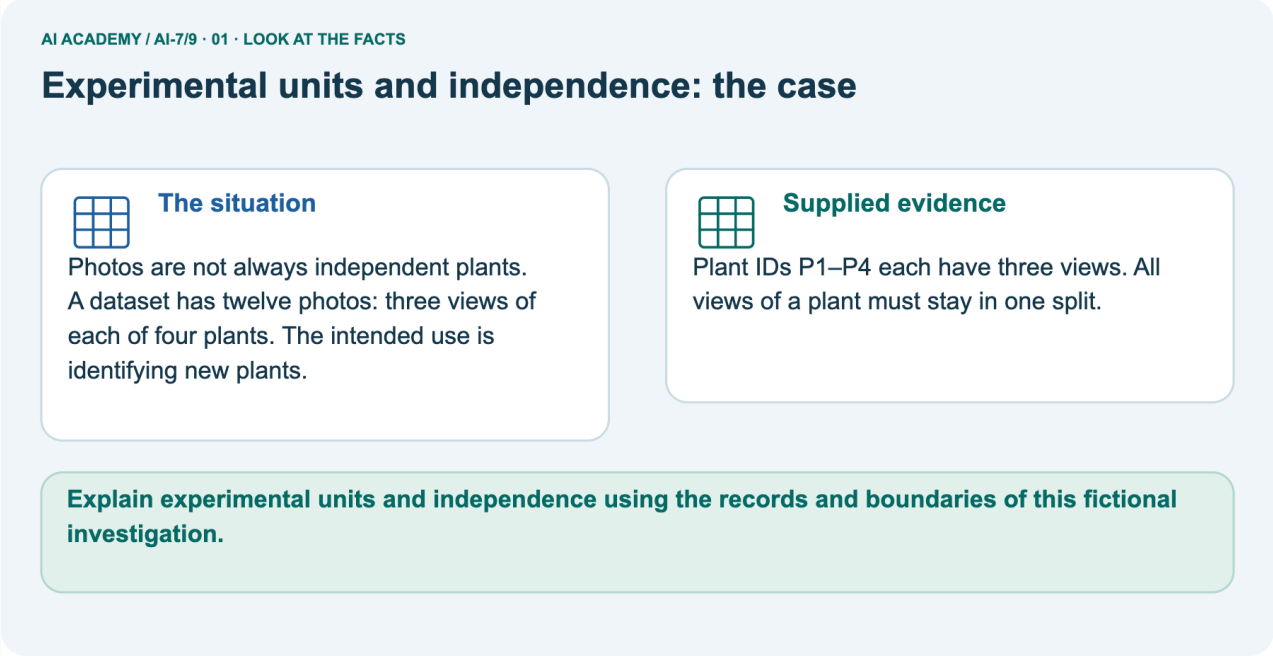
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

# Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

## 01. Experimental units and independence: the case



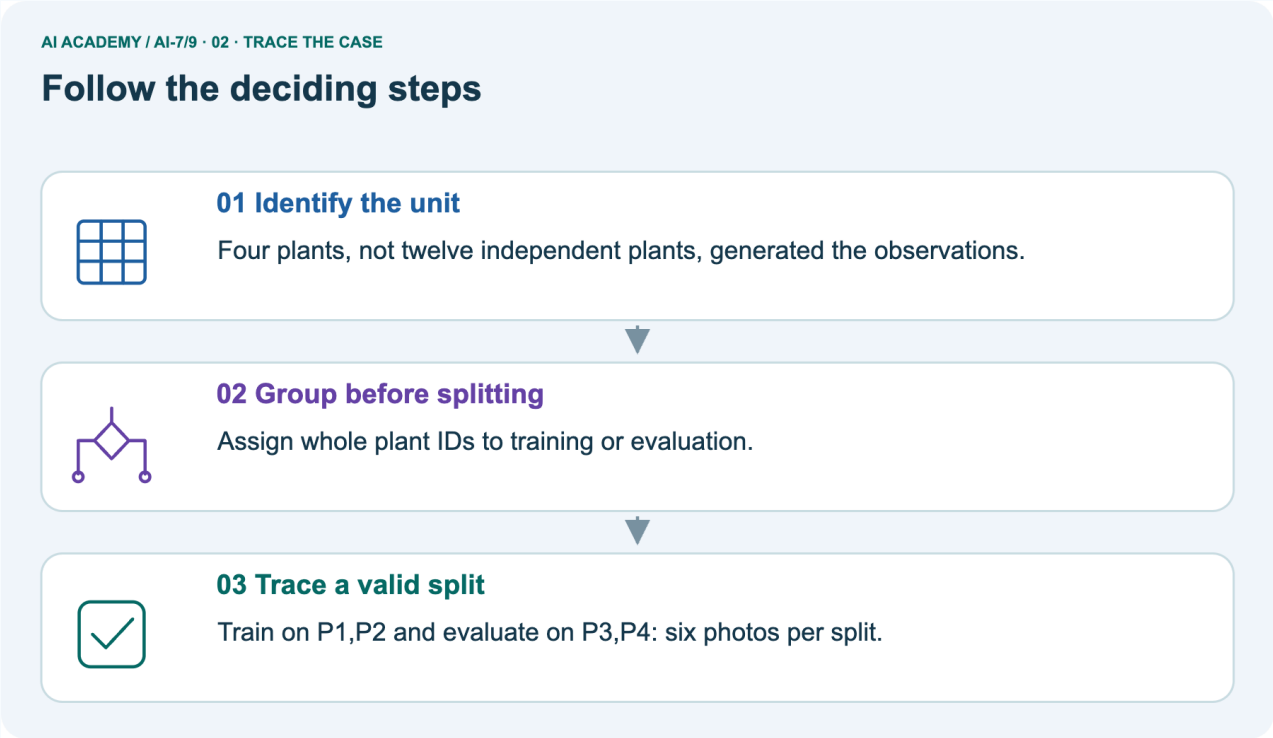

Follow the labelled picture with the matching explanation. Open the vector diagram

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants. Supplied facts: Plant IDs P1–P4 each have three views.

All views of a plant must stay in one split.

Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

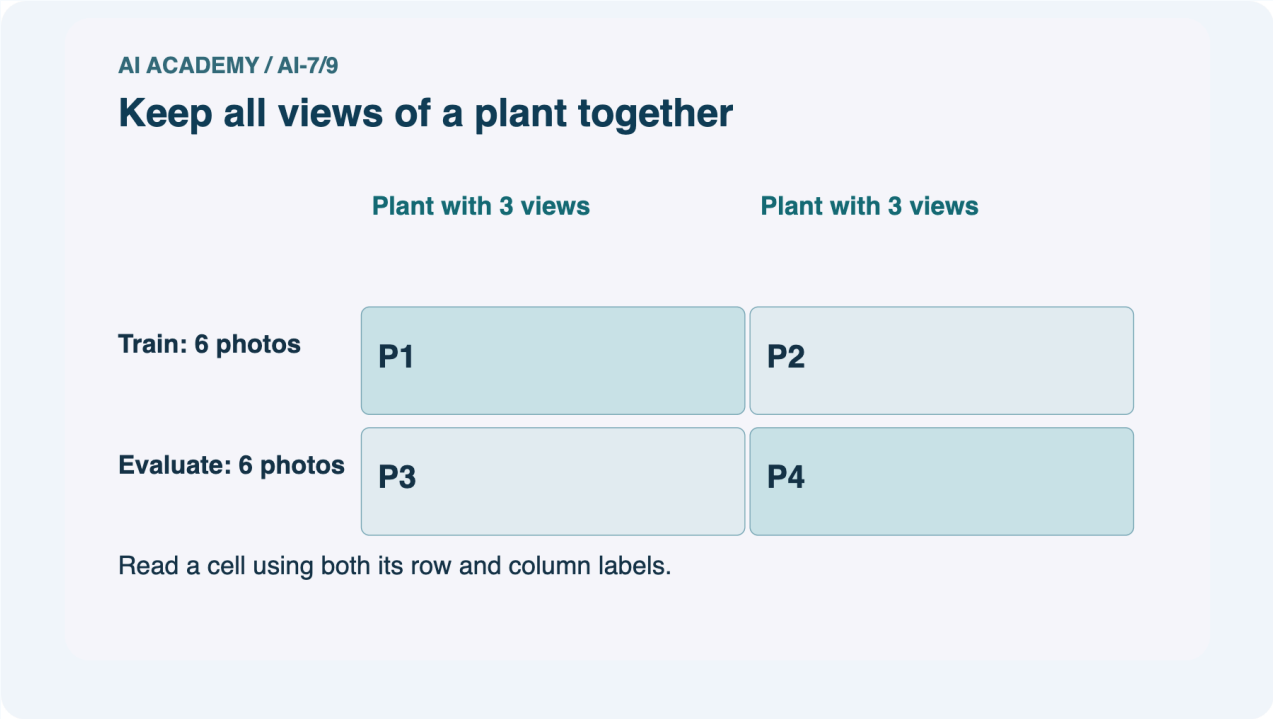



Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the unit: Four plants, not twelve independent plants, generated the **observation**s. → Group before splitting: Assign whole plant IDs to **training** or evaluation. → Trace a valid split: Train on P1,P2 and evaluate on P3,P4: six photos per split.

Point to each numbered stage and explain which recorded **input** it uses.

## 03. Keep all views of a plant together

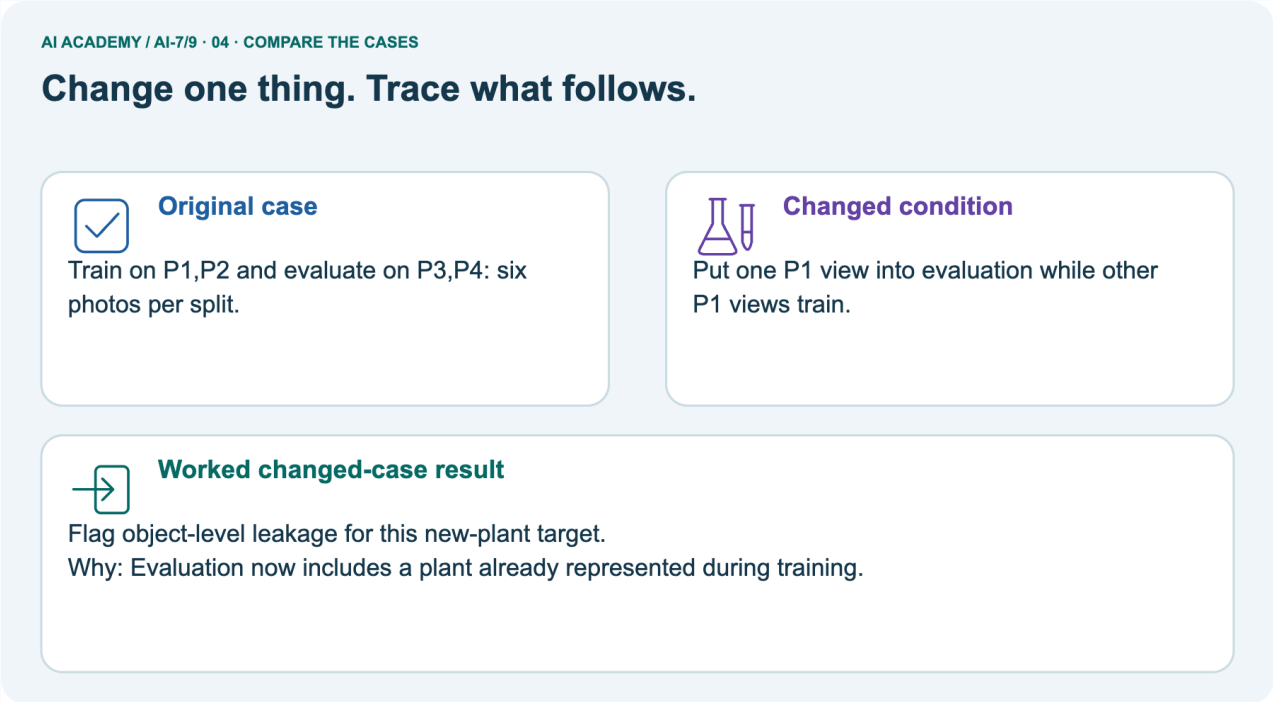



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep all views of a plant together. Train: 6 photos: Plant with 3 views = P1, Plant with 3 views = P2. Evaluate: 6 photos: Plant with 3 views = P3, Plant with 3 views = P4.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



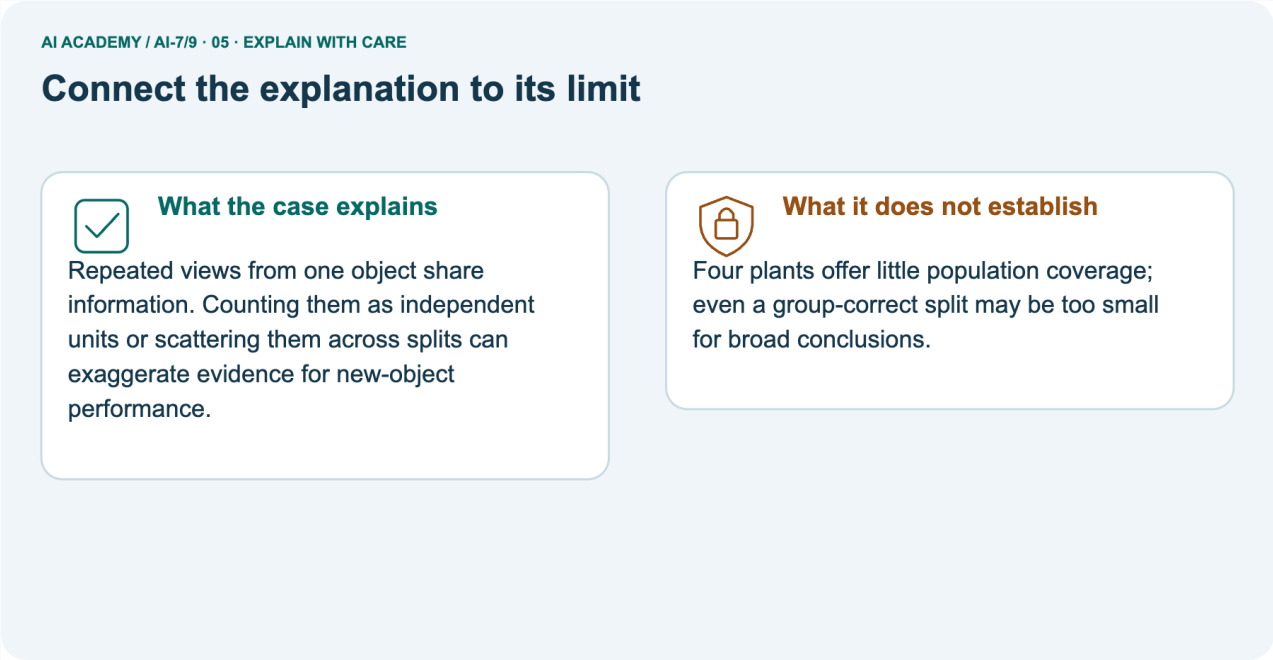

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Train on P1,P2 and evaluate on P3,P4: six photos per split. Changed condition: Put one P1 view into evaluation while other P1 views train. Expected result: Flag object-level leakage for this new-plant target.

Reason: Evaluation now includes a plant already represented during **training**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



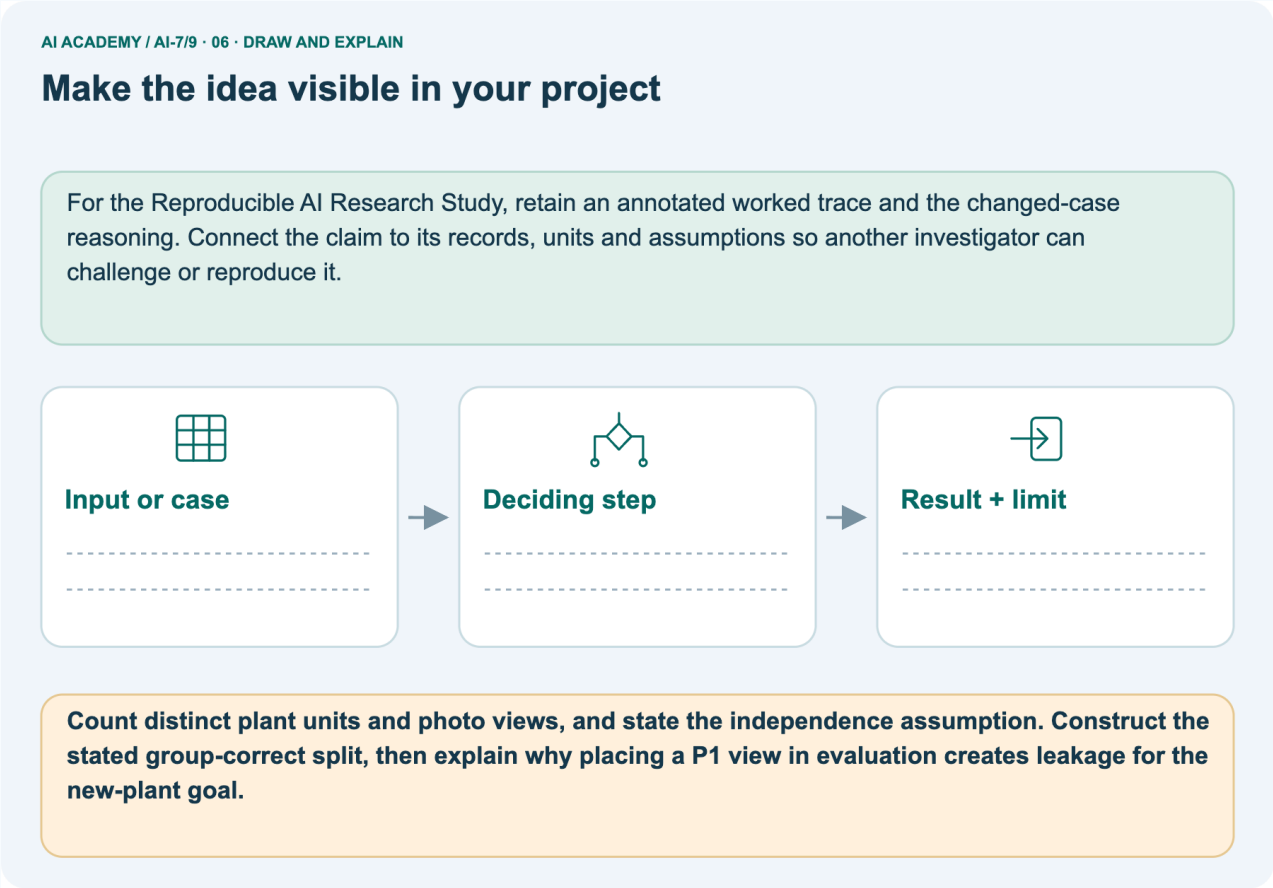

Follow the labelled picture with the matching explanation. Open the vector diagram

Repeated views from one object share information. Counting them as independent units or scattering them across splits can exaggerate **evidence** for new-object performance.

Limit: Four plants offer little population coverage; even a group-correct split may be too small for broad conclusions.

Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it. Draw your **input**, deciding step, result and limit. Supported question: Count distinct plant units and photo views, and state the independence assumption. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

Draw your project trace and justify the deciding step using **evidence**.

## ▼ Earlier wording, alternatives and picture transcripts

### 01. Experimental units and independence: the case

### 02. Follow the deciding steps

### 03. Keep all views of a plant together

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.